EcoLens

Sustainable Environment & Climate · AWS CendekiAwan 2.0

Greener packaging decisions, without the guesswork.

EcoLens helps Malaysian food SMEs pick greener packaging they can afford, from one photo, without hiring a packaging consultant.

Powered by Amazon Bedrock | English & Bahasa Malaysia | Costs in RM | SDG 9 · 12 · 13

Encik Syaful, 44.
Home baker, Kajang.

SELLS Biskut raya in clear plastic jars, through WhatsApp, Shopee and corporate hampers

PEAK About **2,000 jars** in the four weeks before Hari Raya

STAKES One season pays for most of his family's year



Representative persona based on reported Malaysian sellers. Photo: biskut raya jars, Wiki Farazi, CC0.

THE PROBLEM

A client asks for plastic-free hampers. Every option he finds fails him.

Costs more than he can absorb

Kraft tube RM1.90 vs RM0.85 jar. **+RM2,100** across one Raya season.

Doesn't protect the product

Kuih bangkit go soft in our humidity. Paper tubes get crushed by couriers.

He can't tell which claim is real

"Biodegradable" jars nobody in Kajang collects. They'd go to landfill anyway.

No expert, no time

A consultant costs more than his margin. Minimum order 5,000. Ramadan is his busiest month.

Result **He keeps the plastic and loses the hamper order.**

IT'S NOT JUST HIM

Sellers are stuck between cost, durability and recycling reality.



E-commerce sellers: Green packaging costly, less durable

Monday, 30 Jun 2025

Related Stories

NATION 09 Oct 2026
Budget 2027: Govt

The Star, 30 Jun 2025. A kuih chang seller in Ampang: paper wraps don't survive parcel handling, so she stays with plastic.



Bernama, 9 Sep 2025. Food and beverage packaging is "often considered too expensive or too hard to be recycled... most of these items end up in landfills."

1.09 million MSMEs · 96.1% of Malaysian businesses (DOSM 2024)

EPR packaging rules mandatory by 2030

HOW IT'S SOLVED TODAY

Every existing option misses something an SME needs.

	AFFORDABLE	KNOWS HIS PRODUCT	MALAYSIAN RECYCLING	SAME DAY
Packaging consultant	No	Yes	Yes	Weeks
Supplier's advice	Free	Partly	Biased to stock	Yes
Online guides	Free	Generic	Western systems	Yes
EcoLens	RM29 / month	From his photo	Built in	About 1 minute

OUR SOLUTION

One photo in. A supplier-ready decision out.

1

Snap the pack

Photo plus a few details. English or BM.

2

Three options

Quick win, balanced, ambitious. Each with cost, difficulty and risk.

3

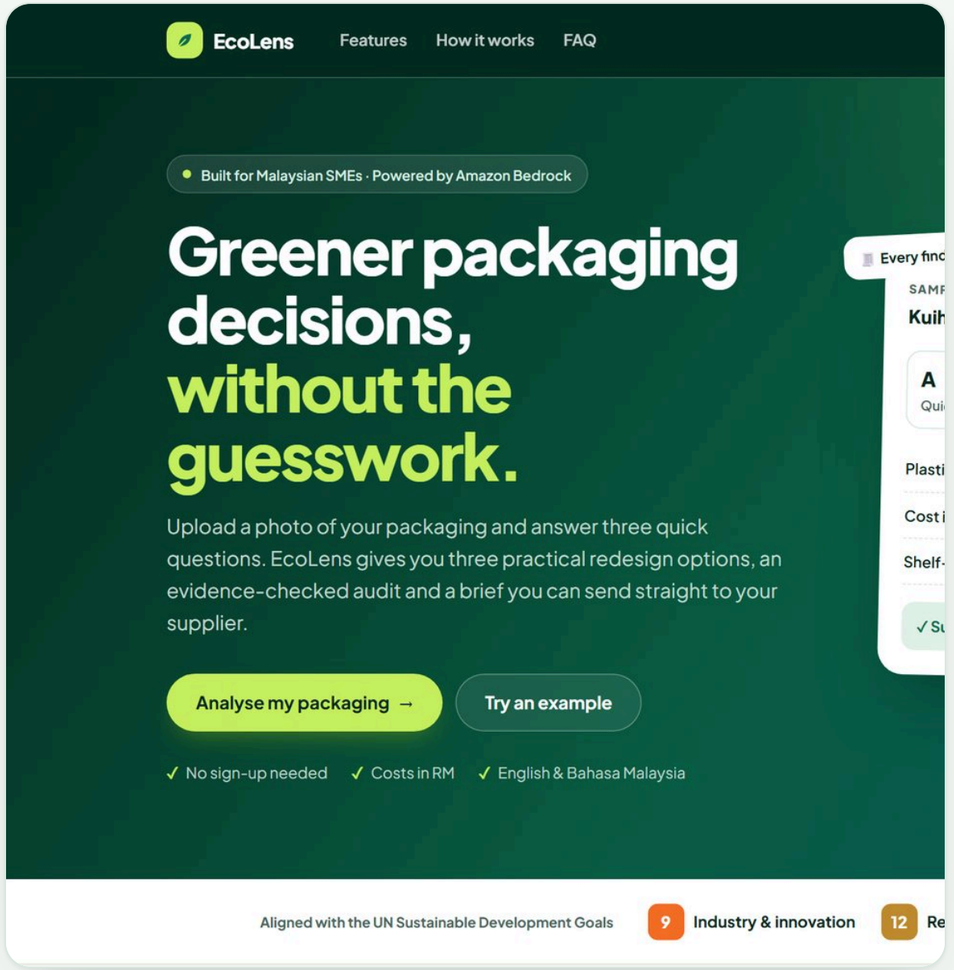
Checked evidence

Every finding labelled Observed, Inferred, Verified or Unknown.

4

Best fit and brief

The option that fits his constraints, and a brief to send his supplier.



The live EcoLens website.

WHY IT NEEDS AI

A form or a search engine can't do these four things.

Vision — Reads his packaging from a photo

Sees a clear jar, a coloured lid, a tray and no resin code, so he **doesn't have to know material names.**

Reasoning — Weighs trade-offs for his product

Balances RM budget, shelf life in our humidity, courier damage and hand-filling, **together, for this product.**

Unstructured input — Turns uncertainty into labels

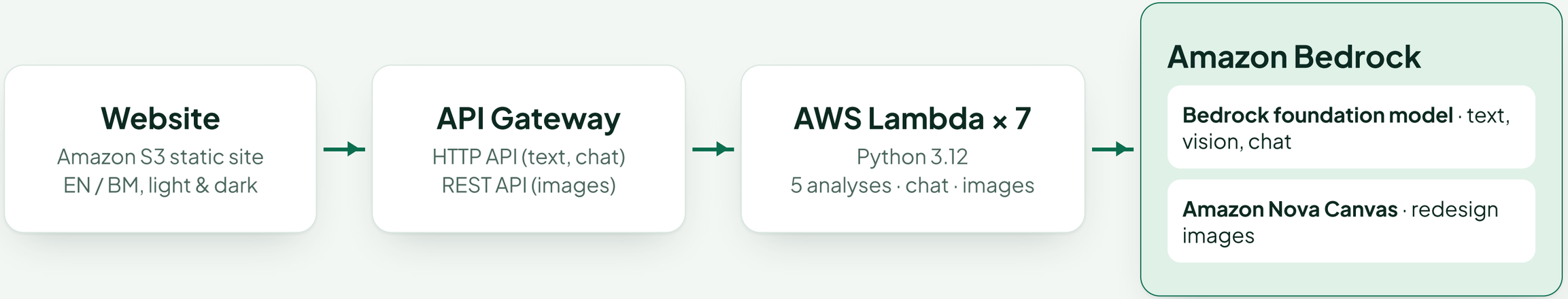
Separates what it can see from what it is guessing, so he **knows what to ask his supplier.**

Language — Speaks his language

Understands English, Bahasa Malaysia and Manglish, and **answers follow-up questions** about his own report.

AWS ARCHITECTURE

Built on the AWS serverless architecture, in the Malaysia region.



Region ap-southeast-5 (Malaysia)

Infrastructure as code: AWS SAM

CI/CD: GitHub Actions on every push

Follows the AWS serverless reference architecture

Built with the Kiro coding agent

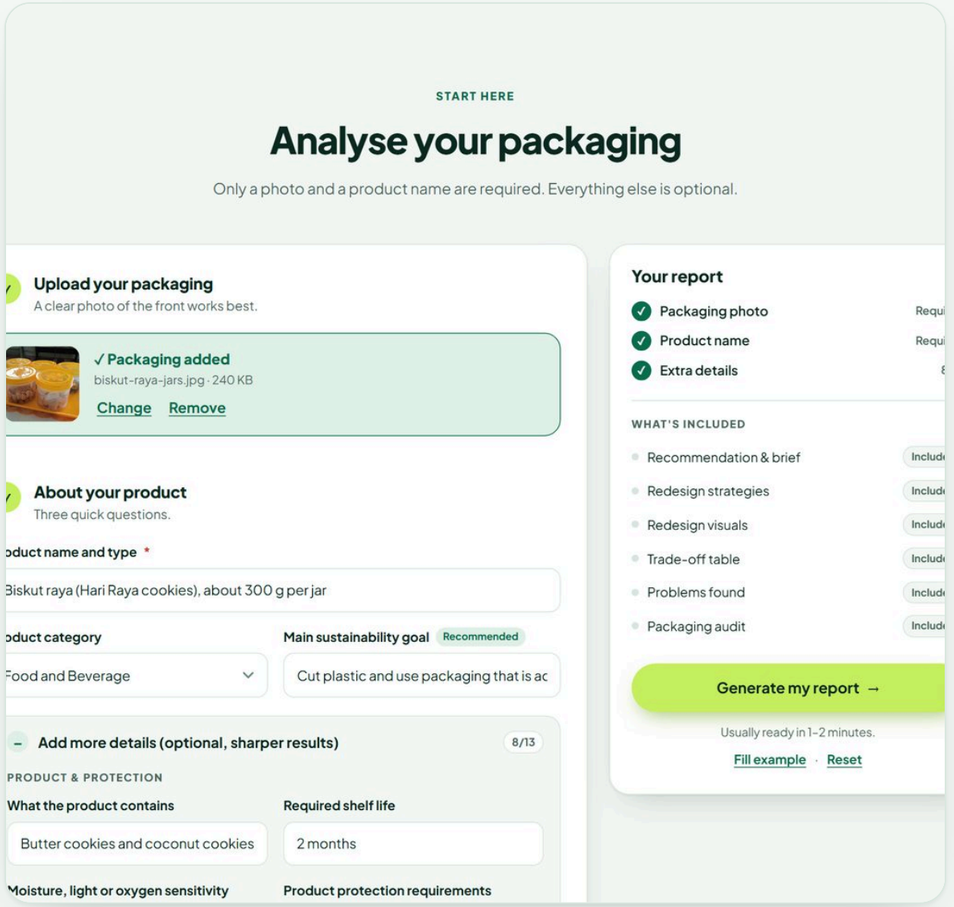
LIVE DEMO

Three real inputs, live on AWS.

1 **Encik Syaful's biskut raya jars**
Photo plus budget (RM0.30 more per jar max) → best-fit option and supplier brief.

2 **Same pack, in Bahasa Malaysia**
Switch to BM and ask the advisor a follow-up question.

3 **A photo that isn't packaging**
EcoLens refuses to invent an answer: "Wrong input detected."

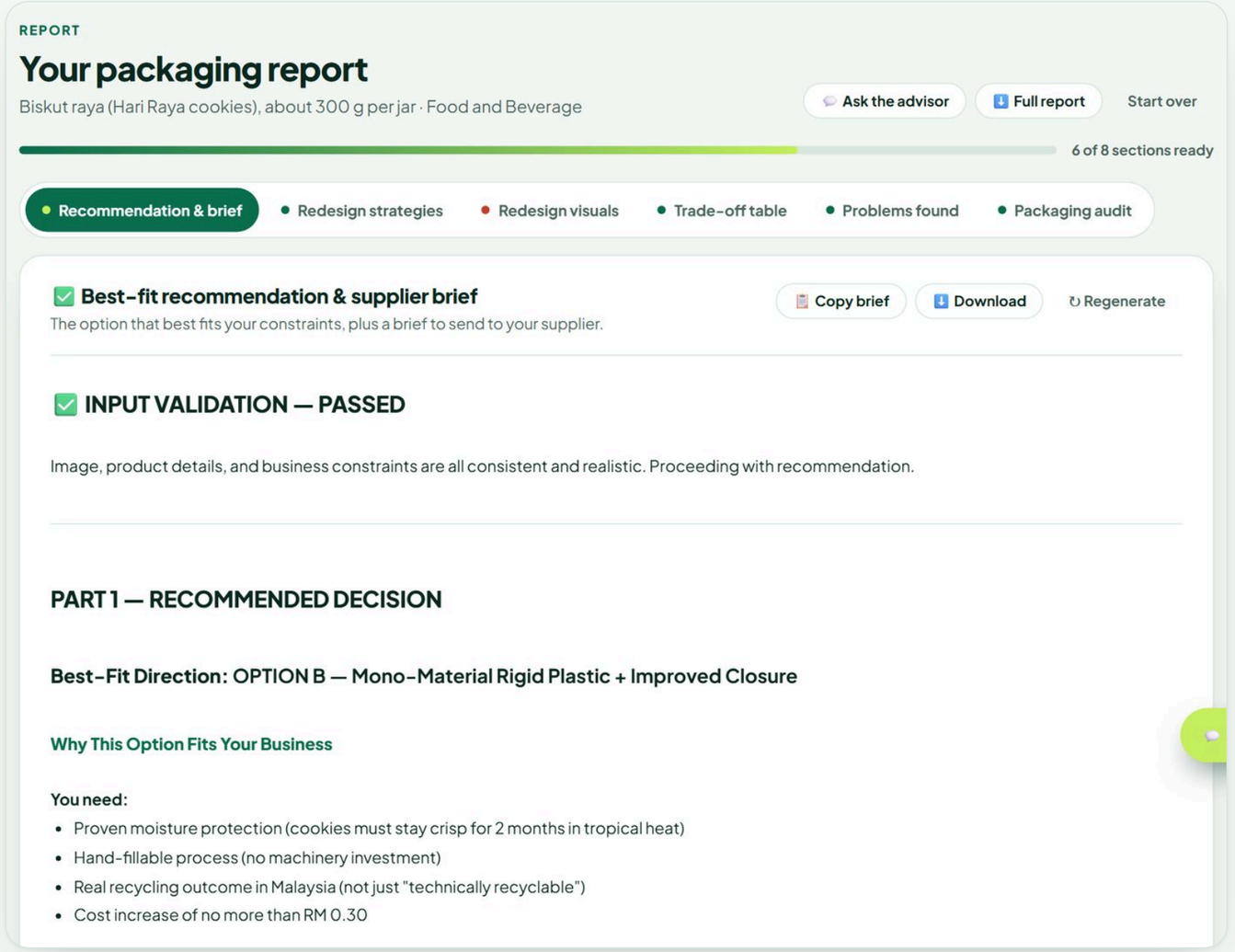


Backup video ready if the venue network fails.

Best fit: keep the jar, fix the material.

EcoLens picked **Option B, a single-material PP jar**: cookies stay crisp, it's hand-fillable, it's "actually collected and sorted in Klang Valley", and it stays within his RM0.30 limit.

Within budget | No new machine | Recyclable in Malaysia



REAL OUTPUT · TRADE-OFFS AND EVIDENCE

Honest about cost, and honest about what it can't see.

COMPARISON TABLE: BISKUT RAYA PACKAGING OPTIONS

Criteria	Current Packaging	Option A: Paper Tube + Desiccant	Option B: Mono-Material PP Jar + Integrated Desiccant	Option C: Cardboard Box + Reusable Glass Jar
Material Reduction	Baseline (3 components)	🟢 High (~60% plastic reduction)	🟢 High (~40% plastic reduction)	🟢 Very High (zero single-use plastic)
Plastic Reduction Potential	100% plastic	🟢 ~40% remaining	🟢 ~60% remaining (mono PP)	🟢 ~5% remaining (only desiccant sachet)
Material Simplicity	🔴 Mixed (PET jar + PP lid + PP tray)	🟢 Simple (kraft paper + minimal PP lid)	🟢 Simple (mono PP jar + lid)	🟡 Moderate (cardboard + glass + beeswax cloth)
Recycling Simplicity	🔴 Hard (mixed plastics, colored tray rarely accepted in Malaysia)	🟢 Easy (kraft paper widely recycled in Klang Valley)	🟢 Easy (mono PP accepted in most Klang Valley schemes)	🟢 Easy (cardboard + glass both recyclable; beeswax compostable)
Product Protection	🟢 Good (rigid jar + tray cushioning)	🟡 Moderate (kraft tube rigid but less tested; requires desiccant validation)	🟢 Good (rigid PP jar, proven for cookies)	🟡 Moderate (cardboard box requires internal padding; glass fragile)
Shelf-Life Risk	Low (proven 2 months)	🟡 Medium (desiccant effectiveness in tropical heat must be tested)	Low (PP + integrated desiccant proven)	🟡 Medium (paper may soften in high humidity; glass hygiene on return)
Cost Impact	RM 0.85 baseline	RM 0.95–1.05 (+RM 0.10–0.20)	RM 1.05–1.10 (+RM 0.20–0.25)	RM 1.20–1.40 upfront (–RM 0.30–0.50 if 50% jars returned)
Within Budget?	✓ Yes	✓ Yes (RM 1.15 max)	✓ Yes (RM 1.15 max)	🟡 Tight (requires deposit model to work)

Trade-off table: cost, protection, shelf-life risk, recyclability in Malaysia.

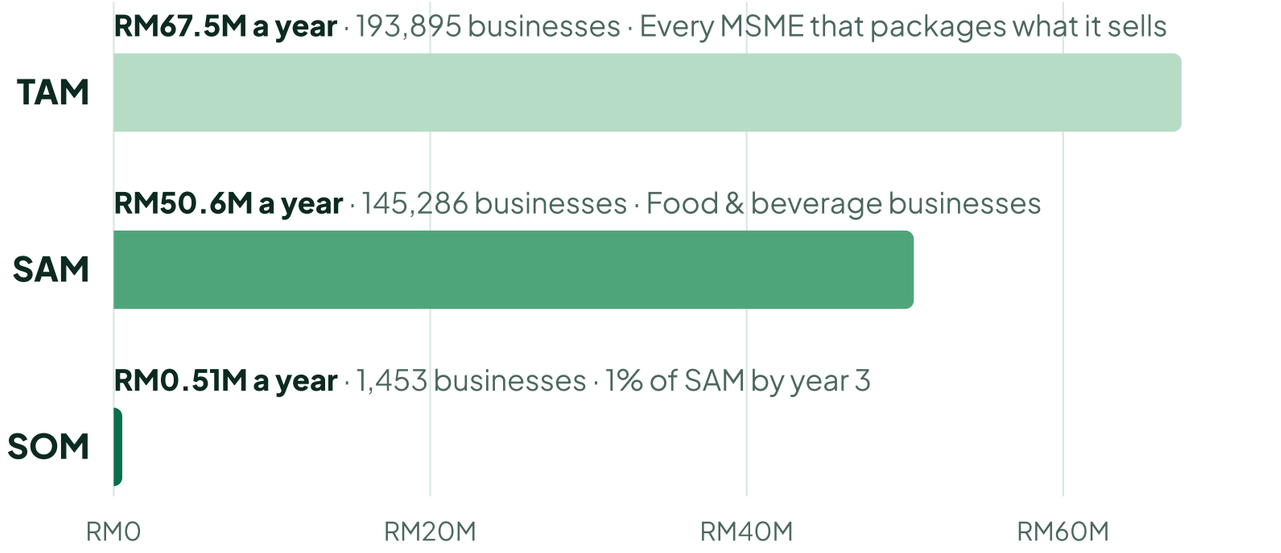
STEP 2 — EVIDENCE CLASSIFICATION

Finding	Classification	Notes
Jar material (PET, HDPE, PP)	**UNKNOWN / REQUIRES VERIFICATION**	No resin code visible. Likely PET or PP based on clarity and rigidity, but must confirm with supplier.
Lid material	**UNKNOWN / REQUIRES VERIFICATION**	Appears to be PP (common for screw caps), but unconfirmed.
Recycled content in jar or lid	**UNKNOWN**	No claim visible; assume virgin plastic unless supplier provides certificate.
Tray material	**UNKNOWN / REQUIRES VERIFICATION**	Likely PP or HDPE; no code visible.
Moisture barrier performance	**INFERRED**	Screw lid + tight seal should keep cookies crisp for 2 months in tropical climate, but depends on jar headspace and seal quality.
Crushability in courier delivery	**INFERRED**	Rigid jar + tray provides moderate protection; cookies inside still at risk if tray is compressed. No additional cushioning observed.
Recyclability in Malaysia	**INFERRED**	Clear PET jars *may* be accepted in some Klang Valley recycling schemes, but uptake is inconsistent. Yellow tray and lids are harder to recycle (colored plastic, lower demand).
Halal compliance	**UNKNOWN**	Packaging itself is halal-neutral, but no halal mark visible on label. Cookies inside must carry halal certification.

Evidence labels: jar material is "Unknown, requires verification", not a guess.

MARKET SIZE

145,000 food businesses decide on packaging every year.



TAM	Manufacturing MSMEs (57,187) + F&B businesses (136,708) × RM348 / year
SAM	Food and beverage businesses: the users our halal, shelf-life and humidity checks are built for
SOM	1% of SAM in 3 years through one pilot partner and SME agencies

DOSM MSME 2024 (via OECD); DOSM Economic Census 2023. Price RM29 / month.

BUSINESS MODEL AND SCALE

Forty sen to run. RM29 a month to use.

RM29

per month, unlimited reports. First 3 reports free.

≈RM0.40

AWS cost per report: 6 Bedrock model calls + 1 image (estimate)

≈94%

gross margin at 4 reports a month per subscriber

RM2,100

wrong-choice cost one decision can save Encik Syaful in one season

What breaks first at 100,000 users

- **Bedrock rate limits** on the text model: request a quota increase and queue bursts.
- **Image service quotas**: move fully to Nova Canvas in us-east-1.
- **30-second API limit**: switch long sections to true streaming.

Who gets left out, and our plan

- **Low digital literacy**: BM by default, photo-first flow, brief sent by WhatsApp.
- **No smartphone or data**: run sessions through SME agencies and co-ops.
- **Not a certification**: EcoLens tells users what to verify with suppliers.

PILOT AND NEXT THREE WEEKS

Pilot with SME Corp Malaysia's food entrepreneurs before Raya 2027.

The pilot

- **50 home-based food sellers**, February 2027, ahead of the Raya rush
- Measure: switches made, cost per pack, supplier briefs sent
- Success: 1 in 3 sellers changes at least one packaging part

If we had 3 more weeks

- **Photo-only flow**: no form, one best-fit answer first (already built on a branch)
- **Stronger images**: Nova Canvas for every redesign visual
- **Local supplier directory** so the brief goes to a real Malaysian supplier

ENVIRONMENTAL IMPACT

One seller, one switch. Multiplied by fifty.

🧁 **ENCIK SYAFUL — HOME BAKER, KAJANG**

32

kg of plastic packaging used per Raya season
(2,000 clear plastic jars × ~16 g each)

2,000	~16 g	32 kg
plastic jars per season	packaging weight per unit	plastic per seller per season

1 **seller switches**
32 kg of packaging can be recycled or composted
🌾 **32 kg saved**

10 **sellers switch**
320 kg diverted — the weight of a small car's fuel tank in plastic 🌾 **320 kg saved**

50 **sellers in our pilot switch**
1,600 kg of packaging recycled every Raya — before we've raised a single ringgit 🌾 **1,600 kg saved per season**

Assumes 1 in 3 sellers fully switches (our pilot success threshold). Aligned with **SDG 12** (Responsible Consumption) & **SDG 13** (Climate Action).

EcoLens

Encik Syaful shouldn't have to choose between his income and the planet. EcoLens shows him the option that protects both.

Our ask: a pilot partner

An SME agency or food co-op to put EcoLens in front of 50 sellers before Raya 2027.

Our ask: AWS credits

To run the pilot and move every image to Nova Canvas.

Why we win

Built for Malaysia: our recycling reality, halal, humidity, BM and costs in RM. Big platforms won't build for a 2,000-jar baker.